1 Theory shape relative to a reasoner

The axioms of a mechanics textbook do not yield the solutions of its exercises by a simple mechanical procedure; the reader brings methods, lemmas and much understanding that is not written down.[1] The formalism of the previous sections cannot express this. It charges a theory by the length of a program for it, a short program can contain a generic solver, and by Craig's construction a consistent assigner becomes an axiom set at constant cost (**??**); so the idealised inducer does not distinguish a theory from a program (**??**). A precise sense in which a written theory is well or badly shaped must be relative to a reasoner that uses it and to a cost that charges what is written. We study the simplest model in which this can be stated. Program length sees neither the reasoner nor the vocabulary. Charged as written, the size of a usable theory depends exponentially on the pair (reasoner, vocabulary) (Table 1). Declarative against procedural form matters only by a constant factor once intermediate concepts are available and the reasoner can chain through them; without them, or with one round of reasoning, it matters for one parity law by exactly the number m of query directions (Theorem 1.13).

1.1 The model

A law constrains the joint values of m observed quantities; a question gives some of the values and asks for another, as in the conditional questions of **??** with literals for sentences; a theory is a set of clauses or equations, which may mention auxiliary variables, the intermediate concepts (theoretical quantities, partial results); a reasoner is a fixed inference procedure.

Definition 1.1 (Theories, laws, questions, reasoners, inducers)**.** Fix base variables $x_1, \dots, x_m$ and auxiliary variables $y_1, y_2, \dots$ (0-ary relation symbols of a first-order language $\mathcal{L}_m$ with equality).

(a) An *explicit theory* Φ is a finite set of non-tautological clauses, a *linear theory* a finite set of equations over GF(2), in these variables; $\operatorname{proj}\Phi := \{x \in \{0,1\}^m : \exists y\ (x,y) \models \Phi\}$. Theories that differ by a renaming of auxiliary variables are identified (the canonical one uses $y_1, \dots, y_{m_{\mathrm{aux}}}$). The classes are $\mathcal{E}_m^{\mathrm{base}}$ (no auxiliary variables) and $\mathcal{E}_m$ (any), and likewise for linear theories. $|\Phi|$ is the number of clauses or equations, $\|\Phi\|$ the number of literal occurrences (the *written length*).

(b) A *law* is a nonempty $\mathcal{S} \subseteq \{0,1\}^m$. A *conditional question* (α, ℓ) is a partial assignment α to the base variables and a literal ℓ on a base variable outside $\operatorname{dom}\alpha$; $\mathcal{Q}(\mathcal{S}) := \{(\alpha,\ell) : \alpha$ is consistent with $\mathcal{S}$ and $\mathcal{S} \wedge \alpha \models \ell\}$. A conditional data set is a finite set of conditional questions.

(c) A *reasoner* $\mathcal{R}$ maps a theory and an antecedent α to a set of base literals or to $\bot$. It is *sound* if $\ell \in \mathcal{R}(\Phi,\alpha)$ implies $\Phi \cup \alpha \models \ell$ and $\bot$ occurs only if $\Phi \cup \alpha$ is unsatisfiable. The named reasoners, all sound: UP_k, k parallel rounds of unit propagation (in each round every clause whose literals are all falsified but one, that one unassigned, assigns it, all such clauses at once; opposite derivations or a falsified clause give $\bot$), and $\mathrm{UP} := \mathrm{UP}_\infty$; GE (Gaussian elimination, on linear theories) and PS_∞ (unbounded proof search), which return exactly the entailed base literals.

(d) Φ is $\mathcal{R}$*-compatible* with D if for every $(\alpha,\ell) \in D$, $\Phi \cup \alpha$ is satisfiable and $\ell \in \mathcal{R}(\Phi,\alpha)$; it is $\mathcal{R}$*-usable* for $\mathcal{S}$ if moreover $D = \mathcal{Q}(\mathcal{S})$ and $\operatorname{proj}\Phi = \mathcal{S}$.

[1] That written theories are far from sufficient for a simple reasoner, and are aids for a reasoner with much tacit understanding, is argued in Hänni's working notes (Hänni, 2026).

(e) The inducers, with hypotheses the linear theories for GE and the explicit theories otherwise: *program length* $K_A^{\mathcal{R}}(D) := \min\{|p| : V(p) = \langle\Phi\rangle,\ \Phi\ \mathcal{R}\text{-compatible with } D\}$, $W_A^{\mathcal{R}}(D) := \sum 2^{-|p|}$ over the same p; *listing* $K_{\mathrm{lst}}^{\mathcal{R}}(D) := \min \ell_{\mathrm{lst}}(\Phi)$, $W_{\mathrm{lst}}^{\mathcal{R}}(D) := \sum 2^{-\ell_{\mathrm{lst}}(\Phi)}$ over the canonical compatible Φ, with $\ell_{\mathrm{lst}} \geq \|\cdot\|$ the prefix code **??** that lists the clauses literal by literal; *time-charged* $Kt_A^{\mathcal{R}}(D) := \min\{|p| + \log t_p\}$ over the p of $K_A^{\mathcal{R}}$, t_p the running time of V on p. The class is $\mathcal{E}_m$; the superscript base restricts it to $\mathcal{E}_m^{\mathrm{base}}$.

For affine laws and for the implication chain on $m \geq 3$ variables, compatibility with $\mathcal{Q}(\mathcal{S})$ already forces $\operatorname{proj}\Phi = \mathcal{S}$ (**??**). Requiring the theory to be consistent with each antecedent, as (E3)does, is necessary:

Proposition 1.2 (The two-valued encoding is degenerate)**.** *Encode (α, ℓ) as the sentence $\alpha \to \ell$ labelled true* (E2)*, with $B = \emptyset$. Every consistent strengthening of a compatible theory is compatible. For every law $\mathcal{S}$ and $x^0 \in \mathcal{S}$ the theory $\{x_i^{x_i^0} : i \leq m\}$ proves every $\alpha \to \ell$ with $(\alpha, \ell) \in \mathcal{Q}(\mathcal{S})$, so $K_A(\{(\alpha \to \ell, 1) : (\alpha, \ell) \in \mathcal{Q}(\mathcal{S})\}) \leq K(x^0) + c$, which is at most $K(m) + c$ for a homogeneous affine law.*

Proof. Compatibility is upward closed. If α agrees with x^0, then $x^0 \in \mathcal{S} \wedge \alpha$ satisfies ℓ, which the theory proves; otherwise it proves $\neg\alpha$. □

Under (E2) the data cannot distinguish a law from a single world that satisfies it. Under (E3), with unbounded proof search, the reasoner-relative inducers are K_A and W_A on question data, up to a constant for r.e. hypotheses and up to $m + O(\log m)$ bits for finite explicit theories (**??**): they extend SAI.

1.2 Program length sees neither the reasoner nor the vocabulary

The prime implicates $\mathrm{PI}(\mathcal{S})$ are the inclusion-minimal clauses over the base variables that hold on $\mathcal{S}$; one round of unit propagation on $\mathrm{PI}(\mathcal{S})$ answers every question of $\mathcal{Q}(\mathcal{S})$ (**??**).

Theorem 1.3 (Reasoner invariance under program length)**.** *Let $\mathcal{R}'$ be a sound reasoner whose hypotheses are explicit or linear theories, of either class. For every conditional data set D, $K_A^{\mathrm{UP}_1,\mathrm{base}}(D) \leq K_A^{\mathcal{R}'}(D) + c$ and $W_A^{\mathrm{UP}_1,\mathrm{base}}(D) \geq 2^{-c} W_A^{\mathcal{R}'}(D)$; and $K_A^{\mathrm{UP}_1} = K_A^{\mathrm{UP}} = K_A^{\mathrm{PS}_\infty} \pm c$ for either class.*

Proof idea (**??**). If Φ' is $\mathcal{R}'$-compatible with D, soundness makes $\mathrm{PI}(\operatorname{proj}\Phi')$ UP_1-compatible with D, and a program for it is a fixed prefix followed by a program for Φ'.

So on the question data of any law one round of unit propagation over the base variables is as good as elimination or proof search with any intermediate concepts. This is Craig's collapse again, for reasoners and vocabularies: the know-how a weak reasoner lacks lives in a generator of the written theory, at constant cost. It becomes visible when the theory is charged as written, or the time to write it is charged. An *affine law* is $\mathcal{S} = \{x : Ax = b\} \neq \emptyset$ with $A \in \mathrm{GF}(2)^{r \times m}$ of rank $r \geq 1$.

Theorem 1.4 (Where the reasoner and the vocabulary become visible)**.** *Let $\mathcal{S}$ be an affine law and $\mathcal{R}$ a sound reasoner; either class.*

(a) Listing: $K_{\mathrm{lst}}^{\mathcal{R}}(\mathcal{Q}(\mathcal{S}))$ *is the least $\ell_{\mathrm{lst}}(\Phi)$ over the $\mathcal{R}$-usable Φ of the class.*

(b) Time-charged: $Kt_A^{\mathcal{R}}(\mathcal{Q}(\mathcal{S})) \geq \log\min\|\Phi\|$ *over the $\mathcal{R}$-usable Φ of the class; $Kt_A^{\mathrm{UP}_1}(\mathcal{Q}(\mathcal{S})) \leq Kt(A,b) + \log|\mathrm{PI}(\mathcal{S})| + O(r + \log m)$, $Kt_A^{\mathrm{UP}}(\mathcal{Q}(\mathcal{S})) \leq Kt(A,b) + O(r + \log m)$ and $Kt_A^{\mathrm{GE}}(\mathcal{Q}(\mathcal{S})) \leq Kt(A,b) + O(\log m)$.*

(c) Program length: $K_A^{\mathcal{R}}(\mathcal{Q}(\mathcal{S})) = K(\mathcal{S}) \pm c$ *for* $\mathcal{R} \in \{\mathrm{UP}_k\ (k \geq 1), \mathrm{UP}, \mathrm{PS}_\infty, \mathrm{GE}\}$.

Proof idea (**??**). (a) Compatibility with $\mathcal{Q}(\mathcal{S})$ is usability (**??**). (b) Printing Φ takes $\|\Phi\|$ steps; a program can compute (A, b) and print the circuit clauses, a circuit library or the equations. (c) $\mathcal{S}$ gives $\mathrm{PI}(\mathcal{S})$, and any compatible Φ gives $\operatorname{proj}\Phi = \mathcal{S}$.

The separations below are therefore stated for written size, which by (a) is K_{lst} up to the code length per literal, with what survives under time charging.

1.3 Over the base variables: the exact minimum

How large must a theory be for unit propagation, the simplest rule-applying reasoner, if it may use only the observed quantities? A *derived equation* of an affine law is $u = yA$ with constant $b_u := y \cdot b$, so $u \cdot x = b_u$ on $\mathcal{S}$; d_* is the least weight of a nonzero one. A *circuit* is a nonzero derived equation whose support contains the support of no other. $C_{Z,\beta}$ is the clause over Z falsified exactly by β; the prime implicates are the $C_{\mathrm{supp}(u),\beta}$ with u a circuit and $u \cdot \beta \neq b_u$ (**??**). A circuit u is *decomposable* if $u = u' + u''$ for nonzero derived equations whose supports meet in one base variable outside $\mathrm{supp}(u)$: two derived equations chained through one intermediate base quantity. Let $\Phi_{\mathrm{ind}}(\mathcal{S}) := \{C_{\mathrm{supp}(u),\beta} : u \text{ an indecomposable circuit, } u \cdot \beta \neq b_u\}$.

Proposition 1.5 (What is forced)**.** *For every affine law, whatever* d_**, the clauses contained in every* UP*-usable explicit theory over the base variables are exactly those of* $\Phi_{\mathrm{ind}}(\mathcal{S})$.

Theorem 1.6 (The unique minimum when $d_* \geq 3$)**.** *If* $d_* \geq 3$*, then* $\Phi_{\mathrm{ind}}(\mathcal{S})$ *is* UP*-usable. Hence an explicit theory over the base variables equivalent to* $\mathcal{S}$ *is* UP*-usable if and only if it contains* $\Phi_{\mathrm{ind}}(\mathcal{S})$*, and* $\Phi_{\mathrm{ind}}(\mathcal{S})$ *is the unique minimum, with* $\sum_{u \text{ indecomposable}} 2^{|u|-1}$ *clauses.*

Proof idea (**??**). From an antecedent fixing all variables of an indecomposable circuit but one, the first round must fire a clause supported on that circuit, since a clause through another variable would decompose it; every other clause can be dropped from the set of all implicates. For sufficiency, induct on the weight: a decomposable circuit is answered in two steps through its intermediate variable, and $d_* \geq 3$ makes both parts lighter.

A circuit must be written down if and only if it cannot be obtained by chaining two derived equations through one base quantity: base quantities already act as intermediate concepts. **??** says when the given equations suffice; for $d_* \leq 2$ the minimum statement fails (**??**).

Corollary 1.7 (The simplex family)**.** *Let* $r \geq 3$*,* $m = 2^r - 1$*, and let the columns of* A *be the nonzero vectors of* $\mathrm{GF}(2)^r$*. Every nonzero derived equation has weight* $(m+1)/2$ *and is an indecomposable circuit, and the unique minimum* UP*-usable theory over the base variables is* $\mathrm{PI}(\mathcal{S})$*, with* $m\,2^{(m-1)/2}$ *clauses of width* $(m+1)/2$*;* GE *uses* $r = \log(m+1)$ *equations.*

For $r = 4$ this is 1,920 clauses against 4 equations. The same holds for one round with any intermediate concepts (Proposition 1.8), but not for unbounded propagation with them (Remark 1.10): the simplex family separates the reasoners exponentially only over the base variables or for one round.

1.4 Intermediate concepts help only a reasoner that can chain through them

One parity law $\mathrm{Par}_m^b := \{x : x_1 \oplus \cdots \oplus x_m = b\}$ needs its 2^{m-1} clauses over the base variables (**??**). With the partial sums $y_1 = x_1 \oplus x_2$, $y_i = y_{i-1} \oplus x_{i+1}$ as intermediate concepts it is written with $4m - 6$ clauses on which unit propagation derives every entailed literal (**??**). Whether this helps depends on how far the reasoner can chain.

Proposition 1.8 (One round cannot use intermediate concepts)**.** *For any law $\mathcal{S}$, every* UP_1*-usable explicit theory, with any auxiliary variables, contains* $\mathrm{PI}(\mathcal{S})$*: 2^{m-1} clauses for one parity law and $m2^{(m-1)/2}$ for the simplex family.*

Proof idea. In one round from an antecedent on the base variables, a clause with an auxiliary literal can derive only that literal; so the base literals come from clauses over the base variables, and **??**(b) applies (**??**). □

Proposition 1.9 (Rounds and size for one parity law)**.** *(a) For $m \geq 4$ the chain encoding is* UP_{m-1}*-usable and not* UP_{m-2}*-usable.*

(b) The balanced-tree encoding*, along a binary tree of height $h := \lceil \log m \rceil$ with the base variables as leaves, an auxiliary $y_v = y_{v'} \oplus y_{v''}$ at each non-root internal node (four clauses) and $y_{v'} \oplus y_{v''} = b$ at the root (two clauses), has $4m - 6$ clauses ($m \geq 2$) and is* UP_{2h-1}*-usable.*

(c) There is an absolute constant $c_{\mathrm{H}} > 0$ such that for every $k \geq 1$ and every m large enough given k, every UP_k*-usable Φ for Par_m^b, with any auxiliary variables, has $3\|\Phi\| + 2m \geq 2^{c_{\mathrm{H}}(m-1)^{1/(2k-1)}}/k$; likewise with $w - 1$ for $m - 1$ for every affine law with a circuit of weight w.*

(d) So UP_k*-usable theories of polynomial size for one parity law need $k = \Omega(\log m / \log\log m)$, and $2\lceil \log m \rceil - 1$ rounds suffice for $4m - 6$ clauses.*

Proof idea (**??**). (a), (b): follow the round in which each auxiliary value becomes known. (c): k rounds of unit propagation are a monotone depth-$2k$ circuit with at most $k(3\|\Phi\| + 2m)$ gates (**??**) that computes parity of $m - 1$ bits, and Håstad's theorem (Håstad, 1986) bounds its size.

Remark 1.10 (The simplex family with intermediate concepts). For the simplex family with $r \geq 4$, writing each circuit (of weight $w = 2^{r-1}$) by its chain encoding gives a UP-usable theory of $m(2m - 4)$ clauses of width at most 3 (**??**); with balanced trees it is $\mathrm{UP}_{2\lceil \log w \rceil - 1}$-usable. So with intermediate concepts and enough rounds the gap to GE is polynomial in written size, and $O(\log m)$ bits under time charging (Theorem 1.4(b)).

1.5 Systems of laws

For a system of laws, intermediate concepts and depth need not help. In the following law the observed quantities are fixed linear functionals of an unobserved state, the natural intermediate concept. Let Λ_q be the set of $(u, c) \in \mathrm{GF}(2)^{q+1}$ with $u \in \mathrm{GF}(2)^q$ of weight 1 or 3, and $\mathbf{e}_\star := (0^q, 1)$. The *hidden-state law* on the $m = |\Lambda_q| + 1$ base variables $(x_\lambda)_{\lambda \in \Lambda_q}$, $x_\star$ is $\mathcal{H}_q := \{((z \cdot \lambda)_{\lambda \in \Lambda_q}, z \cdot \mathbf{e}_\star) : z \in \mathrm{GF}(2)^{q+1}\}$. With $z = (y, z_0)$ as auxiliary variables it is the m equations $x_{(u,c)} = u \cdot y + c\, z_0$, $x_\star = z_0$, of weight at most 5, which GE uses directly.

Theorem 1.11 (Systems of laws need exponential written length)**.** *There are $\varepsilon > 0$ and q_0 such that for every $q \geq q_0$, every* UP*-usable explicit theory Φ for $\mathcal{H}_q$, with any auxiliary variables, clause widths and number of rounds, has $\|\Phi\| \geq 2^{q^\varepsilon/2 - O(1)} - m = 2^{\Omega(m^{\varepsilon/3})}$.*

Proof idea (**??**). From the antecedent setting a set S of the x_λ to 0, $x_\star$ is determined exactly when the XOR system $\{u \cdot y = c : (u, c) \in S\}$ is unsatisfiable. So unit propagation on a usable Φ, as a monotone circuit (**??**), computes this unsatisfiability with $O((\|\Phi\| + m)^2)$ gates; the monotone 3-XOR-SAT function is a projection of it and needs monotone circuits of size 2^{q^ε} (Göös et al., 2019).

$K^{\mathcal{R}}_{\text{lst}}$: $\mathcal{R}=$	parity law, base	parity law, intermediate	simplex family, intermediate	hidden-state law, intermediate
UP_1	2^{m-1} (**??**)	2^{m-1} (Prop. 1.8)	$m2^{(m-1)/2}$ (Prop. 1.8)	$2^{\Omega(m^{\varepsilon/3})}$ (Thm. 1.11)
UP_k, k fixed	2^{m-1} (**??**)	$2^{\Omega(m^{1/(2k-1)})}$ (Prop. 1.9)	$2^{\Omega(m^{1/(2k-1)})}$ (Prop. 1.9)	$2^{\Omega(m^{\varepsilon/3})}$ (Thm. 1.11)
UP_k, $k \geq 2\lceil \log m \rceil - 1$; UP	2^{m-1} (**??**)	$4m-6$ (Prop. 1.9)	$\leq m(2m-4)$ (Rem. 1.10)	$2^{\Omega(m^{\varepsilon/3})}$ (Thm. 1.11)
GE	1 equation	1 equation	$\log(m+1)$ equations	m equations of weight ≤ 5

Table 1: The regimes (Cor. 1.12): the least size of a theory usable by the reasoner $\mathcal{R}$ of the row, with the round budget the row states, in the class of the column (base: no auxiliary variables; intermediate: any). The inducer is the listing inducer $K^{\mathcal{R}}_{\text{lst}}$ (superscript base in the first column), whose value is this size up to the code length per literal (Thm. 1.4). Exact entries count clauses (equations for GE); asymptotic entries bound the written length. Under program length every entry is $K(\mathcal{S}) \pm c$.

Theorem 1.11 combines the monotone-circuit view of unit propagation (Bessiere et al., 2009) with the bound of Göös et al. (2019); it improves the superpolynomial bound of Gwynne and Kullmann (2014) for arc-consistent representations of XOR systems, which by **??** are our usable theories (**??**). We did not find it stated; it is probably known to specialists. Intermediate concepts private to one law add nothing to a system (**??**). A solver whose unit propagation does linear algebra over parity constraints (Beame and Sun, 2026) is a reasoner for which such laws are cheap.

1.6 The regimes

Corollary 1.12 (The regimes)**.** *For one parity law ($m \geq 3$), the simplex family and the hidden-state law, the least size of a theory usable by* UP_1*, by* UP_k *with k fixed, by* UP_k *with $k \geq 2\lceil \log m \rceil - 1$ or* UP*, and by* GE*, over the base variables or with any intermediate concepts, is as stated in Table 1.*

We read the table as follows. The exponential separations are between pairs (reasoner, vocabulary). Intermediate quantities help a reasoner only if it can chain through them: not at all with one round, not enough with any fixed number of rounds, and fully, for one law, from about $2 \log m$ rounds. For the hidden-state law no vocabulary helps unit propagation at any depth, while elimination needs only the natural one.

1.7 Directionality

A natural guess is that logical form matters because a law can be used in every direction (Kepler's third law gives the period from the semi-major axis and conversely), while a program computes one direction. A *directed rule* $P \Rightarrow \ell$ has a set of literals P and a literal ℓ on a variable not in P; it is *sound* for $\mathcal{S}$ if P is consistent with $\mathcal{S}$ and $\mathcal{S} \wedge P \models \ell$, and has written length $|P| + 1$. FC_k is forward chaining in k parallel rounds; usability is defined as for clauses (the projection of a rule set is that of its clauses $\neg P \vee \ell$). A *rule base* is a set of sound rules over the base variables.

Theorem 1.13 (Directionality is real but cheap)**.** *(1) For one parity law on $m \geq 2$ variables, the only* FC*-usable rule base is the set of all $m2^{m-1}$ sound rules, of length m each, against 2^{m-1} clauses of length m for the least* UP*-usable theory over the base variables: a factor m, the number of query directions, in count and in written length. With any auxiliary variables, every* FC_1*-usable set of sound rules contains all $m2^{m-1}$ sound rules over the base variables, against the 2^{m-1} clauses every* UP_1*-usable theory contains.*

(2) With intermediate concepts and unbounded rounds every law has an FC*-usable rule set of written length at most 9 times the least written length of a* UP*-usable clausal theory (3 for the chain encoding of parity). For every round budget k, with or without auxiliary variables, the least* FC_k*-usable rule set has at least as many rules as the least* UP_k*-usable theory has clauses, and at most w times as many, w the largest clause width of a smallest such theory; the same holds for written length. A program with internal variables answers all m directions of parity with $O(m)$ operations.*

(3) Under program length the factor vanishes: $K_A^{\mathrm{FC}_k} = K_A^{\mathrm{UP}_1} \pm c$ on every conditional data set.

Proof idea (**??**). A clause of width w is w directed rules, and FC_k on these rules derives what UP_k derives on the clause. For parity a sound rule concluding x_j must mention all other $m-1$ variables. Splitting wide clauses into clauses of width 3 with fresh variables keeps usability.

The dependence on the number of directions is exactly linear, but it is an artefact of forbidding intermediate concepts or of a single round, and of comparing two tabular forms; a program with internal variables pays nothing for directions. Logical form does not get its support from directionality: the large separations are between pairs (reasoner, vocabulary).

1.8 Where the know-how lives

Proposition 1.14 (Three homes of know-how)**.** *For an affine law, the know-how to answer every question of $\mathcal{Q}(\mathcal{S})$ can be paid for (a)* in the reasoner*: the r equations with* GE*, $O(rm \log m)$ bits; (b)* in written lemmas*: for* UP*, exactly $\Phi_{\mathrm{ind}}(\mathcal{S})$ over the base variables when $d_* \geq 3$, or a circuit library with intermediate concepts (**??**), polynomial for the simplex family and exponential for some systems (Theorem 1.11) and for one parity law at fixed depth; (c)* in uncharged computation*: in a generator under program length (Theorem 1.3), or in a membership procedure with free search for derivations. For the latter, for a suitable polynomial q,* $\mathrm{PI}(\mathcal{S})$ *has a decider of length $K^q(A,b)+c$ running in time polynomial in m, and for every $(\alpha,\ell) \in \mathcal{Q}(\mathcal{S})$, $\bigwedge \alpha \to \ell$ follows from one member of* $\mathrm{PI}(\mathcal{S})$ *by propositional reasoning; $K^q(A,b) \leq rm + r + O(\log m)$, and $K^q(A,b) = O(\log m)$ for one parity law and the simplex family.*

That these one-instance derivations have size polynomial in m, with a degree independent of m, we only sketch (**??**). Charged deduction (**??**) extends to conditional data (**??**); a hypothesis is a decider with the degrees of its time and derivation budgets.

Theorem 1.15 (Charged deduction does not charge for the lemma library)**.** *Let $\mathcal{S}$ be an affine law and take any charged-deduction reading of* **??***, at polynomial budgets with degrees charged. (i) If each derivation of Proposition 1.14(c) has size at most $(|\bigwedge \alpha \to \ell| + 2)^{j_0}$, the decider of* $\mathrm{PI}(\mathcal{S})$ *with derivation degree j_0 is compatible with $\mathcal{Q}(\mathcal{S})$ and has weight at least $2^{-K^q(A,b) - O(\log m) - 2\lceil \log(j_0+1) \rceil}$. (ii) Every hypothesis compatible with $\mathcal{Q}(\mathcal{S})$ has weight at most $2^{-K(\mathcal{S})+c}$.*

Proof idea (**??**). (i) counts the decider's length and degrees. (ii) A compatible hypothesis determines $\mathcal{S}$: $x \notin \mathcal{S}$ if and only if some $\bigwedge x|_I \to \ell$ with ℓ false at x has a derivation within the budget, which is decidable.

With $j_0 = O(1)$, as we sketch, these readings charge the full lemma library ($m2^{(m-1)/2}$ clauses for the simplex family) within $K^q(A, b) - K(\mathcal{S}) + O(\log m)$ bits of the least possible cost. They measure the length of a membership test, not of what a reader must be handed; the test for the library runs elimination, so the know-how lives in the decider. This locates the fit classes of **??**: the readings bound the size of a derivation that exists and do not charge finding it (**??**), so they model a reasoner that is handed the derivation, and cannot express the textbook point, which concerns a reasoner that must find the answer. A bounded deterministic reasoner with a cost on the written theory can.

1.9 Learning, and depth against size

A weak reasoner must also acquire its know-how (**??**). For one parity law each question fixes $m-1$ variables; its *pattern* is the non-solution obtained by flipping the asked variable in the completion, and there are 2^{m-1} patterns of m questions each. Without intermediate concepts, every inclusion-minimal UP-compatible theory (for instance a shortest one under the listing code) fails on some question of every pattern not yet seen, so it is fully correct only after all 2^{m-1} patterns have been seen; on data that contain whole patterns the listing inducer pays at least $\mathrm{len}_m \approx m \log m$ bits per pattern, in the semimeasure convention. With GE, or with intermediate concepts and unbounded propagation, the listing inducer pays $O(m \log m)$ bits in all; under program length there is no gap.

Depth trades against size even without new vocabulary (**??**). For the implication chain $x_1 \to x_2 \to \cdots \to x_m$, a 2-clause theory over the base variables is UP_k-usable if and only if its edges form a k-hop transitive-closure spanner of the path, and the least UP_k-usable theory over the base variables, with clauses of any width, has exactly as many clauses as a minimum such spanner, which is $\Theta(m\,\mathrm{ia}_k(m))$ (Alon and Schieber, 1987; Bhattacharyya et al., 2012): $\binom{m}{2}$ at $k = 1$, $\Theta(m \log m)$ at $k = 2$, $\Theta(m \log\log m)$ at $k = 3$, down to the $m-1$ axioms of the law at $k \geq m-1$. At depth 2, intermediate concepts and wide clauses save at most a factor of about 2 (**??**).

1.10 What this shows and does not show

In first-order logic the gap between a lemma-free theory and the same theory with its lemmas can be non-elementary for a cut-free reasoner (Statman, 1979; Orevkov, 1982); valid lemma sentences, which change the written theory but not its content, restore short cut-free proofs, as we sketch in **??**. Whether the picture extends beyond Boolean laws is open:

Question 1.16. Does the dependence of written size on the pair (reasoner, vocabulary) extend beyond Boolean laws: to real-valued laws with numerical solvers, to probabilistic programs with conditioning, to simulation codes?

This is the precise form of the open question why theories are shaped like logical theories (**??**). Our reading of the results is this. Whether a written theory suffices depends on the reader. Paired with a reasoner that can do algebra, a linear law's few equations suffice. Paired with one that only applies rules, the theory must spell out exponentially many derived laws, unless it is written with new intermediate quantities and the reader can chain many steps through them. Even then, some systems of laws need exponentially long written theories for such a reader, whatever quantities are introduced. The idealised inducer cannot see any of this, because it charges a theory by the length of a program that may compute the missing lemmas. The model does not show that logic, rather than probabilistic programs with conditioning or classes of models, is the right formal home for theories: the large separations are between pairs (reasoner, vocabulary), and a rule base

with intermediate concepts is within a constant factor of a clausal theory. It does not touch the gap between a problem as posed and its formalisation, nor tacit skill (**??**). Its laws are linear or implicational, its reasoners are unit propagation, forward chaining and elimination, and the cost of a theory is its written length. It is not a theorem about textbooks.

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